Beyond Recall: A Layered Diagnostic for Write-Time Memory Governance in Language Agents

Ali Mohammadi

Abstract

Persistent language agents can silently corrupt future behavior when they write an ambiguous update, retain a stale value, or replace a record without history. Existing end-to-end question-answering evaluations conflate this write decision with storage, retrieval, and answer generation. We formulate memory writing as KEEP, lineage-preserving SUPERSEDE, or write-time ABSTAIN, and introduce a layered harness that scores the action, resulting state, and later answer separately. We study the harness on MGP-6 v2, an author-designed 180-item pilot diagnostic with typed and provenance-tracked states. The held-out test results do not establish method superiority: a transparent reference policy reaches 0.8889 action accuracy, versus 0.9000 for a structured LLM policy, and their family-clustered difference includes zero. More importantly, a conflict-type-only lookup fitted on train and development reaches 1.000 test accuracy (type-only: 0.9667), demonstrating that the current pilot leaks its labels and cannot isolate evidence reasoning from scenario recognition. The reference policy has zero observed action false commits on 33 test abstention cases, but the one-sided 95% upper bound is 0.0868 and its missed-update rate is 0.2222; under an update-responsive cost matrix the structured LLM is preferable. We release the per-item predictions, shortcut audits, cluster bootstrap, risk–coverage and state/lineage metrics. The contribution is therefore an auditable evaluation protocol and a measured negative finding about diagnostic design, not a claim of general memory-governance performance.

1 Introduction

A language agent that persists across sessions must decide what to do with each candidate memory: retain the current record, replace it with a newer one, or decline to commit because the evidence is insufficient. Deployed systems often make this decision through opaque LLM tool calls (Packer et al., 2023; Chhikara et al., 2025) or additive event logs (Park et al., 2023). A wrong write can remain latent until much later: an ambiguous statement becomes a durable fact, a stale preference survives, or a correction erases its predecessor. End-to-end QA then reveals an error without identifying which layer caused it.

We take the write transition itself as the object of evaluation. The action is KEEP, lineage-preserving SUPERSEDE, or write-time ABSTAIN over typed, provenance-tracked records. This combines established ideas from selective prediction (Chow, 1970; Geifman and El-Yaniv, 2017), belief revision (Alchourrón et al., 1985), and agent memory operations (Yan et al., 2025; Chhikara et al., 2025; Wang, 2026). We claim neither the individual operations nor abstention, typing, provenance, and lineage as new. The contribution is a harness that makes their decision-, state-, and answer-layer consequences separately observable.

We first exercise the harness on MGP-6 v2, a 180-item, English, synthetic pilot. The collection was authored by the same project that implemented the reference policy and has only one independent human re-annotation of its test split. It is therefore a diagnostic instrument, not an independently established benchmark. Rather than treating a held-out item split as sufficient, we explicitly test whether scenario metadata predicts the action and whether item-level inference understates within-template dependence.

Our contributions and findings are:

1. We formalize the three write actions and release a layered evaluator for action correctness, state invariants, lineage, stale retrieval, selective risk, and asymmetric write costs.
2. We add train/dev-fitted shortcut baselines and template-family/minimal-pair cluster bootstrap. Conflict type alone predicts every held-out test action; type alone predicts 96.7%. This is direct evidence that the current pilot does not identify general evidence reasoning.
3. On the held-out test split, a transparent reference policy reaches 0.8889 accuracy and a structured LLM policy reaches 0.9000; the paired family-cluster interval includes zero. Zero observed reference-policy false commits has a non-zero finite-sample upper bound and trades against missed updates, so the preferred policy changes with the cost matrix.

This reframing turns a weakness into a falsifiable result: an item-level test split can look clean while a family label still determines the answer. The released artifacts make that failure visible and specify the gate for a successor collection: independent construction, all three actions within each conflict family, blinded multi-human annotation, trajectory-level cases, and a hidden test set frozen after policy development.

2 Related Work

We position KSA against five clusters. For each we name the nearest precedent and state precisely what we build on. Our claim concerns the operationalization and controlled evaluation of a particular combination, not priority for any individual component.

Agent memory systems with write operations. MemGPT/Letta (Packer et al., 2023), Mem0 (Chhikara et al., 2025), A-MEM (Xu et al., 2025), and RMM (Tan et al., 2025) manage a persistent store with per-item operations (add/update/delete/no-op, or merge-vs-add). Memory-R1 (Yan et al., 2025) is the clearest prior formalization of memory writes as a discrete action space, learned by RL. We do not claim to introduce the action space. The differences we defend are that (i) `NOOP` in these systems means "nothing new," not "too ambiguous to commit," so there is no uncertainty-driven refusal; and (ii) they are primarily rewarded at the answer level, without a shared diagnostic that separately scores the write and its resulting state. Our released rule is a transparent reference policy, not a claim of a new learning algorithm.

Write-time conflict resolution and lineage. Zep/Graphiti (Rasmussen et al., 2025) invalidates contradicted graph edges with bi-temporal timestamps rather than deleting them — the strongest shipped prior on supersession-with-history. TOKI (Wang, 2026) argues, formally, that contradiction resolution *is* write-time concurrency control, and types the heuristic family (including an *await-confirmation* operator functionally equivalent to ABSTAIN) as bitemporal operators with audit rows. FadeMem (Wei et al., 2026) classifies new-vs-old memory pairs and applies decay/fusion. Relative to these, storing lineage or provenance is not new; our distinct piece is to make the write a *governed, scored* decision with an abstain option, and to measure lineage integrity separately from the action label. Reddy and Challaram (2026) independently report that deterministic conflict resolution beats LLM judgment on a consolidation benchmark, motivating independent tests of transparent policies rather than validating our author-designed pilot by analogy.

Typed memory. MemGuard (Ha et al., 2026) assigns each memory an explicit functional role at write time to prevent contamination, and MemOS (Li et al., 2025) types memory levels. Typed memory is therefore precedented; what we add is a measured ablation of typing inside the pilot, reported as a mechanism diagnostic rather than an independently identified causal effect.

Diagnostics and evaluation for memory. HaluMem (Chen et al., 2025) is the nearest prior on operation-level evaluation, separately scoring extraction, update, and QA for hallucinations; we do not claim to introduce operation-level evaluation. STALE (Chao et al., 2026) probes *frozen agents at read time* for recognition of stale beliefs and shares MGP-6 v2's construct names (implicit conflict, premise resistance), but defines no write actions, abstention, or lineage. The Supersede environment (Patel, 2026) isolates the answer-level "supersession gap" on LongMemEval-M and trains the behavior end-to-end. GateMem (Ren et al., 2026) uses the phrase "memory governance" for authorization across principals — a different axis from our epistemic write correctness. LongMemEval (Wu et al., 2025) and LoCoMo (Maharana et al., 2024) are end-to-end QA benchmarks that cannot attribute a failure to the write decision. MGP-6 v2 operationalizes joint scoring of write-time KEEP/SUPERSEDE/ABSTAIN decisions and resulting memory state; we do not claim that operation-level or state-level evaluation itself is new.

Selective prediction, belief revision, knowledge editing. The reject option (Chow, 1970; El-Yaniv and Wiener, 2010; Geifman and El-Yaniv, 2017) and learning to defer (Madras et al., 2018; Mozannar and Sontag, 2020) are the origin of ABSTAIN; RSCB-MC (Iscan, 2026) already applies abstention to memory *retrieval* in coding agents. We therefore never claim to introduce abstention to memory systems; we measure what an explicit abstain-on-write option contributes within our controlled diagnostic and claim no priority for the mechanism. AGM belief revision (Alchourrón et al., 1985) is the intellectual ancestor of KEEP/SUPERSEDE; we cite it as ancestry only and prove no postulate compliance. Knowledge editing (Meng et al., 2022, 2023) governs *how* to change a parametric belief and assumes the edit should happen; we govern *whether* to commit an external write. Finally, PrefEval (Zhao et al., 2025) and PersonaMem (Jiang et al., 2025) motivate testing preference changes on independently constructed natural dialogue.

Scope of contribution. Each component has named precedents: Memory-R1 for write actions; Zep and TOKI for retained history; MemGuard and MemOS for typing; deterministic policy work including Reddy and Challaram (2026); and HaluMem and STALE for evaluation. We therefore make the narrower contribution claim that this work operationalizes and audits an explicit three-way policy — KEEP, SUPERSEDE, or ABSTAIN — over typed, provenance-tracked records with supersession lineage, and evaluates that implementation at three separate layers: decision, stored state/lineage, and downstream answer. The controlled conjunction, its validity checks, and the measurements they enable are the contribution; update operations, version history, provenance, typing, abstention, and operation-level evaluation are not claimed as inventions.

3 Problem Formulation

We formalize write-time memory governance for a single atomic semantic slot, which is the unit MGP-6 v2 controls. This scoping is deliberate: it isolates the write decision from retrieval and answer generation so that the decision can be scored on its own.

3.1 Memory records, state, and types

A memory record m is a tuple $m = (id, \tau, v, c, \sigma, \pi, t, s)$, where id is a unique identifier, $\tau \in \mathcal{T}$ is the memory *type*, v is the stored value (written in the auditable form `slot = value`), $c \in [0, 1]$ is a stored confidence, $\sigma \in \{\mathsf{active}, \mathsf{superseded}\}$ is the record's *status*, π is the supersession pointer (`supersedes_id`, possibly null), t is a validity timestamp, and s is the record's *provenance* (the source event/turn that established it). We use three types, ordered by expected durability:

$$\mathcal{T} = \{\mathsf{stable_pref},\ \mathsf{episodic_fact},\ \mathsf{transient_pref}\}.$$

A *memory state* M is a set of records for the slot. A well-formed state has at least one active record and, for a preference slot, exactly one active stable_pref record; every non-null π points at a retained superseded record.

3.2 Candidate update and evidence

At a write step the agent extracts a *candidate update* u (a proposed new value with a type guess) from a *perturbation event* against the current state M. The decision is driven by *evidence* associated with u: the extraction confidence, the specificity and recency of the supporting span, the presence of negation or hedging markers, and the classified logical relation of u to the active record (support, contradiction, or independence). These evidence signals are the inputs to the policy of Section 5; the formulation itself is agnostic to how they are computed.

3.3 Action space and transition function

The controller emits one action $a \in \mathcal{A} = \{\text{KEEP}, \text{SUPERSEDE}, \text{ABSTAIN}\}$, which induces a deterministic transition $M' = \delta(M, u, a)$:

- $\delta(M, u, \text{KEEP}) = M$ for durable records: the prior active record is preserved unchanged. A *clear temporary* override may be added only as a separate active transient_pref record, never by mutating the durable one.
- $\delta(M, u, \text{SUPERSEDE})$ marks the prior active record superseded and adds one active successor whose π points back at it. The prior value is *retained*, not deleted, and no longer active in the slot; exactly one active record of the successor's type remains.
- $\delta(M, u, \text{ABSTAIN}) \equiv M$ at the modeled field level: the state is byte-for-byte equivalent. ABSTAIN is a decision to *not commit* the candidate, distinct from committing a null value.

A transition is an *invalid commit* if it changes the active state on an item whose gold action is ABSTAIN, or otherwise violates a state-transition invariant (Section 4). This is the failure the abstention and provenance machinery is designed to prevent.

3.4 Four distinctions the formulation makes explicit

These distinctions are load-bearing and are frequently conflated in memory systems; we make them explicit because MGP-6 v2 scores each.

KEEP vs. retrieval absence. KEEP is a positive write-time decision to retain the current active value in the presence of a candidate update; it is not the same as a record simply not being retrieved at answer time. KEEP operates on the state; retrieval absence operates on a later read.

ABSTAIN vs. no-op. A no-op (e.g. Mem0's NOOP) fires when the candidate carries *nothing new*. ABSTAIN fires when the candidate carries something potentially relevant but the evidence is *too weak or ambiguous to commit* (unreliable source, non-atomic claim, referential uncertainty). Both leave the state unchanged, but only ABSTAIN is an uncertainty-driven refusal; this is why we report an external NOOP→ABSTAIN mapping only as an upper bound (Section 7).

Write-time vs. answer-time abstention. ABSTAIN in our formulation is a *write-layer* non-commit: the store is not modified. It is distinct from an answer-layer refusal to respond (as scored by end-task benchmarks), which occurs downstream and does not change memory. MGP-6 v2 withholds the evaluation query until the write is committed precisely so that answer-time behavior cannot leak into the write decision.

Supersession vs. deletion. SUPERSEDE replaces the active value while *preserving* the prior record as superseded with a lineage edge. Deletion (and active forgetting) removes information and is explicitly out of scope for this work; the reserved forgotten status never occurs in our data.

3.5 Objective

Let y be the target action and let $C(a, y)$ be an application-specific cost. The policy objective is

$$\pi^* = \arg\min_{\pi} \ \mathbb{E}[C(\pi(M, u), y)], \tag{1}$$

subject to state and lineage invariants. The cost is intentionally not assumed symmetric: committing on ambiguous evidence, missing a valid update, and unnecessarily abstaining can have different consequences. We therefore report the full action confusion and expected loss under multiple pre-declared cost regimes rather than treating accuracy as a universal utility. State validity, lineage integrity, stale retrieval, and calibration remain separate outcomes; they are not folded into a post-hoc scalar chosen after seeing test results.

4 MGP-6 v2: A Decision-Level Diagnostic

MGP-6 v2 is an author-designed diagnostic slice for write-time memory governance. We use the word *diagnostic* deliberately and throughout: it is a pilot instrument for probing decision mechanisms, not a community benchmark, an independent test collection, or a measure of general QA or retrieval efficiency. All scenarios are synthetic, English-only, and low-risk, and contain no real personal data.

4.1 Conflict taxonomy and distribution

The pilot contains 180 scenarios, 30 for each of six scenario families, with an action imbalance of 90 SUPERSEDE, 66 ABSTAIN, and 24 KEEP that we handle by reporting macro-F1 and per-action results rather than accuracy alone.

Type	Diagnostic condition	Rows	Gold action(s)
T1	preference drift over time	30	30 SUPERSEDE
T2	episodic factual correction	30	30 SUPERSEDE
T3	stable-preference replacement	30	30 SUPERSEDE
T4	ambiguity / referent uncertainty	30	30 ABSTAIN
T5	transient capture trap	30	24 KEEP, 6 ABSTAIN
T6	confidence laundering	30	30 ABSTAIN

T1–T3 target supersession under different durability regimes; a subset of 10 rows per type is *cue-free* (implicit_conflict), i.e. the perturbation contains none of the explicit change cues ("now," "instead of," "actually," ...) yet still fixes a mutually exclusive current value. T4/T6 require ABSTAIN because evidence is referentially uncertain or laundered through an unreliable source; T5 separates a durable preference (KEEP) from a temporary override. Some supersession cases also have minimal-pair T4 partners. These pairs test a narrow cue change, but they do not remove the global shortcut: conflict type still determines the action throughout v2.

4.2 Query modes

Every scenario carries one query mode used only at the answer stage: *direct recall* (108 items), *premise resistance* (45; the query embeds the stale/unsupported value as a false presupposition that a correct answer

must reject), and *implicit application* (27; a concrete task solvable only by using the active value, with neither value named in the query). These modes let us measure whether a system that makes the right write decision also refuses stale premises and applies the active value in context.

4.3 Splits

Splits are 60 train / 30 dev / 90 test, stratified so each type contributes $10/5/15$ rows and T5 contributes $8/2$, $4/1$, $12/3$ KEEP/ABSTAIN. No template-family identifier and no minimal-pair group crosses a split boundary, which blocks literal family overlap. It does *not* block construct leakage when the same family-to-action rule holds in every split. We therefore treat the 90-item test split as a held-out item split only, never as independent validation.

4.4 Pre-specified validity checks

Before comparing policies we fit lookup baselines on train+dev and evaluate them on test using only (i) the six-way type, (ii) the finer conflict type, and (iii) source/reliability/temporal metadata. We also compute macro-accuracy and paired bootstrap intervals by `template_family_id`; for `minimal_pair_group_id`, unpaired items are singleton clusters. A shortcut baseline that approaches the full policy is a failed construct-validity check, not a competitive result.

4.5 Schema, invariants, and scoring

Each row provides a dialogue prefix, prior memory state, perturbation event, source/temporal metadata, a gold action, an auditable gold memory state, the query mode, the expected answer, and a regex judge specification with required (`include`) and forbidden (`not_include`) spans. Evaluation is two-stage: the write stage sees only the prefix, prior state, and perturbation and commits an action and predicted state *before* the query is revealed; the answer stage then answers from the predicted state. A validator enforces the state-transition invariants of Section 3.3 (one active record per slot, retained superseded predecessors, single-successor lineage, byte-equivalent ABSTAIN states, active-only answers), and a released corpus prints `VALIDATION PASSED: 180 examples, 0 errors`.

We score two families of metric (full definitions in Appendix D): *decision-level* — action accuracy, per-action and macro-F1, and the action false-commit rate on gold-ABSTAIN items — and *memory-state* — state invalid-commit on gold-ABSTAIN items, five-condition lineage integrity on SUPERSEDE items, and stale-retrieval rate (the fraction of answers surfacing a forbidden superseded value). Two shipped forgetting-rate metrics saturate at 1.0 for every system and are excluded from all tables as uninformative.

4.6 Annotation: one independent annotator plus a model audit panel

We are precise about the annotation evidence, which has two separate components and must never be described as multi-annotator human agreement.

One independent human annotator re-labelled the 90-item test split against the annotation manual. Raw agreement with gold is 0.9889 (CI_{95} $[0.9667, 1.0]$), Cohen's $\kappa = 0.98146$ (CI_{95} $[0.93997, 1.0]$), with exactly one disagreement (`MGP6V2-T3-TEST-016`, an `implicit_conflict` SUPERSEDE $\rightarrow$ABSTAIN case). This is $n{=}1$ annotator; the annotator's human status rests on author attestation, not on evidence in the released files.

A four-model audit panel (four distinct LLMs) independently labelled the same 90 items: Fleiss $\kappa = 0.9815$ across the panel, with the majority vote matching gold on all 90 items (88 unanimous). This is a model audit, not human inter-annotator agreement.

The larger human review sheet (180 rows) exists but is unfilled; a genuine multi-annotator human release is future work (Section 8). The panel and the single annotator may share the manual's reading because the labelling policy is unambiguous by construction — a construct-validity caveat we carry explicitly.

4.7 Construction grounding

Scenario *shapes* were informed by inspecting real corpora — LongMemEval knowledge-update items and its false-premise (`_abs`) questions, Memora operation chains, GateMem's machine-readable judge structure, and named confidence-laundering patterns — but every released row is a newly authored synthetic paraphrase; no source content is reused. This grounds the failure modes without importing personal data, and it is the reason MGP-6 v2 shares construct names (implicit conflict, premise resistance) with concurrent read-side work (Chao et al., 2026), which we cite to disambiguate.

5 Reference Policies

The name KSA simply abbreviates the three actions (KEEP/SUPERSEDE/ABSTAIN); it is not a claim that the action vocabulary is novel. The released pilot includes two policies over this vocabulary and two shortcut probes. All commit actions pass through the same deterministic state transition and invariant validator.

5.1 Transparent reference policy

The reference policy parses the raw perturbation text and prior state with auditable lexical predicates. In priority order it (1) ABSTAINs on unattributed third-party, laundered, garbled, or unconfirmed evidence; (2) ABSTAINs on a hedged temporary candidate; (3) KEEPs the durable value and optionally stores a separate transient note for a clear temporary override; (4) SUPERSEDEs when a usable durable candidate conflicts with the active value; and (5) otherwise ABSTAINs. The complete predicates and value parser are released in `benchmarks/mgp6/systems/`.

This policy is deterministic and inspectable, but it was written against the same annotation manual and phrase generator as the pilot. It is therefore a *reference implementation of the policy*, not an independently learned method. A five-feature weighted scorecard exists in the broader prototype code, but it did not generate the reported MGP-6 predictions; we do not describe or claim it as the evaluated method. This distinction removes an earlier implementation–manuscript mismatch.

5.2 State transition and lineage

On SUPERSEDE the transition marks the prior record superseded, appends one active successor, and points its `supersedes_id` to the retained predecessor. On KEEP a clear temporary override may be a separate transient record; on ABSTAIN the modeled state is unchanged. The evaluator scores these state properties independently of the action, so a correct action with corrupted lineage is not counted as a correct state transition.

5.3 Structured and unstructured LLM policies

We also run the governance decision through a structured LLM policy prompted with the typed state, candidate, evidence fields, and the three action definitions, and an unstructured zero-shot policy that receives the raw event and prior state without that scaffold. Both use `qwen/qwen3.5-9b`. Because the prompt, candidate representation, and pilot were author-designed together, even this comparison remains diagnostic.

5.4 Shortcut probes

The type-only and conflict-type-only probes learn a majority action for each metadata value on train+dev and apply the lookup unchanged to test. They never read the dialogue, prior value, or candidate evidence. Their purpose is to test whether the collection rewards scenario recognition without memory reasoning.

6 Experimental Setup

6.1 Systems

The main test comparison contains the transparent reference policy, the structured LLM policy, and the unstructured zero-shot policy described in Section 5. Before interpreting them, we run type-only, conflict-type-only, and source/reliability/temporal lookup probes fitted on train+dev. Four structurally limited systems (no memory, flat retrieval, recency, and naive overwrite) and component ablations are retained as pilot mechanism checks in the appendix; they are not competitive peers.

A six-demonstration LLM decision run produced $174/180$ predictions. Aligning its six missing identifiers to `INVALID` yields a coverage-aware audit, but the run is incomplete and excluded from every comparison and hypothesis test.

Adapter runs for Mem0, Letta, Graphiti, and MemOS are preserved as harness diagnostics in the artifact. Their native operations do not match our action vocabulary, so mapped action accuracies are excluded from the main empirical claim. A successor evaluation must compare observable final state, stale retrieval, downstream answer, latency, and cost under each system's native API.

6.2 Metrics and denominators

Primary decision-level metrics are computed over the held-out 90-item test: action accuracy, macro-F1, and per-action precision/recall/F1. We report action false-commit rate over gold-ABSTAIN items, missed-update rate over gold-SUPERSEDE items, and unnecessary-abstention rate over non-ABSTAIN items. A zero event count is accompanied by a finite-sample upper bound.

Memory-state outcomes are lineage integrity (a five-condition per-item score on gold-SUPERSEDE cases) and stale-retrieval rate. We additionally report top-label ECE/Brier, risk–coverage, and expected loss under balanced, commit-averse, and update-responsive cost matrices. Confidence results are descriptive because neither policy was calibrated on an independent set. Full formulas and commands are in Appendix D.

6.3 Statistical methods

Systems are paired by scenario id. The primary uncertainty estimate is a $10{,}000$-replicate bootstrap over `template_family_id`; we repeat it with minimal-pair groups plus singleton unpaired items. We report family-macro and item-weighted estimates. Exact two-sided McNemar tests are secondary because item independence is doubtful. No equivalence or non-inferiority claim is made: no margin was pre-specified.

For zero observed action false commits we report the exact one-sided 95% upper bound $1 - 0.05^{1/n}$ rather than interpreting the observed rate as population zero. These calculations quantify uncertainty inside this collection; they do not repair its construct leakage or establish external generalization.

Annotation agreement is reported as Cohen's κ for one independent human annotator against the author gold and, separately, Fleiss κ for a four-model audit panel. The latter is never described as human inter-annotator agreement.

Table 1: Shortcut probes fitted on train+dev and evaluated on the held-out test split. These probes never read the dialogue or memory state.

Probe	Test accuracy	Macro-F1	Unseen values
Six-way type only	0.9667	0.9471	0
Conflict type only	1.0000	1.0000	0

Table 2: Held-out test results (n=90). Family CIs resample the 48 template families and report family-macro accuracy. Action false-commit is over 33 gold-ABSTAIN items; missed-update is over 45 gold-SUPERSEDE items. The shortcut result in Table 1 limits interpretation of all rows.

Policy	Accuracy	Family 95% CI	Macro-F1	Action false-commit	Missed-update
Reference policy	0.8889	[0.8125, 0.9688]	0.9145	0.0000	0.2222
Structured LLM	0.9000	[0.8333, 0.9688]	0.9040	0.0606	0.1556
Unstructured zero-shot LLM	0.6778	[0.5625, 0.7917]	0.6542	0.5758	0.2222

7 Results on MGP-6 v2

All results below are recomputed from released per-item predictions. We lead with validity checks because they determine how the policy scores may be interpreted.

7.1 The shortcut check fails

The conflict-type-only lookup fitted on train+dev reaches 1.0000 accuracy and macro-F1 on test; the coarser six-way type lookup reaches 0.9667 accuracy and 0.9471 macro-F1. Neither sees the dialogue or memory state. Thus the held-out item split does not test whether a system reasons about evidence quality, provenance, or temporal scope: scenario recognition is sufficient. This failed check dominates every policy comparison below.

7.2 Held-out policy comparison

On test, the reference policy reaches 0.8889 accuracy/0.9145 macro-F1 and the structured LLM reaches 0.9000/0.9040. The reference-minus-LLM risk difference is -0.0111; the paired template-family cluster interval $[-0.094, 0.063]$ includes zero (exact item-level McNemar: $b = 4, c = 5, p = 1.0$). These data support neither superiority nor non-inferiority. The unstructured zero-shot policy reaches 0.6778/0.6542, but that gap is itself evaluated on the shortcut-leaking pilot and is not a general scaffold-effect estimate.

7.3 Safety–responsiveness trade-off

The reference policy makes no action false commits on the 33 test abstention items, but the exact one-sided 95% upper bound is 0.0868, not zero risk. It misses $10/45$ required updates (0.2222) and unnecessarily abstains on $10/57$ non-abstention items (0.1754). The structured LLM action false-commit rate is $2/33 = 0.0606$, while its missed-update rate is $7/45 = 0.1556$. Consequently the reference policy has lower expected loss under the commit-averse matrix (0.111 vs. 0.189), but higher loss under the update-responsive matrix (0.556 vs. 0.433). Safety is therefore cost-conditional, not an absolute ordering. Top-label ECE is 0.156 for the reference confidence and 0.302 for the LLM confidence; both are descriptive because neither confidence was calibrated on an independent set.

Table 3: Within-generator component ablations of the reference policy on MGP-6 v2 (n=180). Each removal produces a distinct diagnostic signature. Invalid-commit is over gold-ABSTAIN items; lineage integrity is over gold-SUPERSEDE items. Values transcribed from results-ledger version 1.1; these are not independent effect estimates.

Configuration	Action acc.	Macro-F1	Induced failure signature
Full system	0.9222	0.9399	invalid-commit 0.0; lineage integrity 0.8556
− abstention	0.5556	0.4278	invalid-commit $0.0 \rightarrow 1.0$
− provenance	0.7000	0.7428	invalid-commit $0.0 \rightarrow 0.606$
− typing	0.7889	0.5680	KEEP F1 $\rightarrow 0.0$
− lineage	0.9222	0.9399	action unchanged; lineage integrity $0.8556 \rightarrow 0.3689$

7.4 State and component diagnostics

Full-corpus ablations are retained only as within-generator mechanism checks. Removing abstention raises invalid-commit from 0.0 to 1.0; removing provenance yields 0.606; removing typing drives KEEP F1 to 0.0. Removing lineage leaves action accuracy unchanged at 0.9222 but drops lineage integrity $0.8556 \rightarrow 0.3689$. The last result demonstrates the central measurement point: action accuracy alone cannot reveal a corrupted transition history.

The $174/180$ six-shot run remains incomplete and is excluded from all comparisons. Mapped external-system action scores and non-homogeneous downstream runs remain in the artifact audit only; neither supports a main-paper capability claim.

8 Discussion, Threats to Validity, and Reproducibility

8.1 What the evidence supports

The evidence supports the usefulness of separating write, state, and answer failures and the necessity of testing diagnostic shortcuts before interpreting policy scores. It also supports within-pilot mechanism observations, such as lineage corruption being invisible to action accuracy. It does not support population accuracy, deployment safety, reference-policy superiority, non-inferiority, or downstream benefit.

8.2 Threats to validity

Internal and construct validity. The same project designed the taxonomy, manual, generator, gold labels, and reference policy. Conflict type perfectly predicts test actions, so item holdout and high annotation agreement cannot establish construct validity. Family-cluster bootstrap corrects one form of dependence but cannot repair the deterministic family–label mapping.

External validity. MGP-6 v2 is small, synthetic, English-only, single-step, and single-slot. It lacks independent generation, natural dialogue, multi-step trajectories, multiple model families, multiple seeds, and native external-system comparisons. None of its rates estimates a deployment population.

Annotation. There is one independent human re-annotation of the test split plus a four-model audit panel. Agreement may be high because the manual and family-to-label mapping are deterministic. The model panel is not human inter-annotator agreement, and one human annotator cannot establish multi-annotator reliability or construct validity.

Statistical conclusion. There are only 48 test template families, and cluster intervals are correspondingly wide. The reference–LLM comparison is a null, not evidence of equality. Confidence calibration and cost matrices were not validated on independent data, so they expose trade-offs rather than certify risk. The one-sided upper bound attached to zero action false commits prevents a sample zero from being presented as a safety guarantee.

8.3 Ethics and scope

The pilot contains synthetic, low-risk scenarios with no real personal data. Our target is epistemic write correctness. Memory poisoning, multi-principal access control, privacy, deletion compliance, and security governance are separate threat models that we do not test. We make no production-safety claim.

8.4 Successor-dataset gate

A successor release must be constructed after the evaluated policies and thresholds are frozen; use an independent generator and hidden labels; include KEEP, SUPERSEDE, and ABSTAIN within every conflict family; add adversarial minimal pairs and 5/10/20-step trajectories; obtain blinded labels from at least three humans; and include natural or independently paraphrased dialogue. The test set must remain sealed until all development choices are fixed. These are requirements for stronger claims, not future-work decorations.

8.5 Reproducibility statement

The dataset, schema, validator, reference policies, and per-item predictions are released. The construct-validity script regenerates shortcut tables, family/minimal-pair cluster intervals, calibration, risk–coverage, and cost sensitivity from those artifacts. The incomplete six-shot run and mapped external adapters remain explicitly non-primary. Reproduction commands are in Appendix E. Independent construction, multi-human annotation, and trajectory data are release gates for a successor benchmark rather than boxes silently marked complete here.

9 Conclusion

We separated memory writing from the state and answer layers it affects, then used that separation to audit an author-designed pilot rather than merely report its highest score. The central empirical result is negative and useful: conflict type alone perfectly predicts the pilot's held-out actions, while the transparent reference policy and structured LLM are statistically unresolved and exchange safety for responsiveness under different costs. The release makes these limitations reproducible through per-item predictions, shortcut probes, cluster-aware uncertainty, finite-sample risk bounds, and lineage metrics. A credible successor must be independently constructed, contain all actions within each conflict family, use blinded multi-human labels, and test multi-step natural trajectories. Until then, MGP-6 v2 is a pilot for evaluating the evaluator, not evidence of general memory-governance performance.

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A Dataset Schema

Each MGP-6 v2 row is a JSON object. Exact field names are grouped below; the machine-readable schema remains authoritative.

```
identity: scenario_id, version, type, subtype, difficulty,
          split, quality_flags
dialogue: base_dialogue_prefix, initial_memory_state,
          perturbation_event, semantic_slot
evidence: evidence_source, source_reliability, temporal_scope,
          conflict_type, memory_age
gold:     gold_action, gold_memory_state
answer:   query_mode, evaluation_query, expected_answer,
          forbidden_answer_patterns, judge_spec
audit:    rationale_notes, annotation_rule_ids, template_family_id,
          minimal_pair_group_id
```

A memory record contains the following fields:

```
id, type, text, confidence, status, supersedes_id, valid_from,
semantic_slot, source
```

The text field uses the auditable `slot = value` form; confidence lies in $[0, 1]$; and status is active or superseded. In v2, the judge's `not_include` list equals the forbidden-answer-pattern list. A correct answer must satisfy every include regex and no not-include regex. Scenario identifiers have the form `MGP6V2-TYPE-SPLIT-NNN`.

State-transition invariants (enforced by the validator). Every state has ≥ 1 active record; a preference slot has exactly one active stable_pref; every supersedes_id points to a retained superseded record; SUPERSEDE retains the predecessor and adds one active successor pointing back at it; KEEP preserves the prior durable record (a clear temporary override may be added only as a separate active transient_pref); ABSTAIN leaves the state byte-for-byte equivalent at the modeled field level; answers contain an active value and no superseded or forbidden value; duplicate equivalent active records are invalid.

B Annotation Process

The annotation manual encodes ten decision rules (R1–R10): require direct-user evidence else abstain (R1); require same user/slot/topic (R2); require an atomic self-contained claim (R3); durable lexical change cues (R4); a reliable durable same-slot contradiction supersedes (R5); a clear temporary override keeps durable memory while a hedged temporary candidate abstains (R6); low-confidence/garbled/ laundered evidence abstains (R7); expected answers use active memory only (R8); preserve superseded records for lineage (R9); no duplicate active memories for paraphrases (R10). Cue-free rows do not depend on R4 and follow inferential certainty under R1–R3 and R5.

Annotation evidence has two components (Section 4.6), reported separately and never conflated: (a) **one independent human annotator** over the 90-item test split — raw agreement 0.9889 (CI $[0.9667, 1.0]$), Cohen's $\kappa = 0.98146$ (CI $[0.93997, 1.0]$), macro-F1 0.9913, one disagreement (MGP6V2-T3-TEST-016); and (b) a **four-model audit panel** over the same items — Fleiss $\kappa = 0.9815$, majority vote matching gold on all 90 (88 unanimous). The 180-row human review sheet is unfilled; a multi-annotator human release is future work.

C Prompts

The extraction and relation-classification prompt templates are released under prompts/; baseline answer prompts are packaged under src/ksa/baselines/prompts/. The combined prompt set is frozen; its SHA-256 (computed as sha256 over the sorted files, each contributed as relpath + NUL + bytes + NUL) is recorded in artifacts/manifests/freeze_hashes.json and reproduced in Appendix E.

D Metric Formulas

Action canonicalization (KEPT→KEEP, NOOP→ABSTAIN, unknown→INVALID) is applied to both sides before comparison; gold actions are only KEEP/SUPERSEDE/ABSTAIN, so INVALID/INVALID_COMMIT predictions are always incorrect.

Action accuracy $\frac{1}{N}\sum_i \mathbf{1}[\,\text{canon}(g_i) = \text{canon}(p_i)\,]$; the primary denominator is the $N = 90$ held-out test split.

Macro-F1 unweighted mean of per-action F1 over {KEEP, SUPERSEDE, ABSTAIN }; an action's F1 is 0 if precision+recall= 0.

Action false-commit rate over gold-ABSTAIN items, $\frac{1}{|\text{ABS}|}\sum_{i\in\text{ABS}} \mathbf{1}[\,p_i \neq \text{ABSTAIN}\,]$; lower is better. This is the denominator used in Table 2 and the cost analysis.

State invalid-commit rate over gold-ABSTAIN items, the fraction for which the predicted state contains an active record outside the allowed gold state. This separate state-level diagnostic catches a nominal ABSTAIN that still writes memory and is used in the full-corpus ablations.

Missed-update rate fraction of gold-SUPERSEDE items not predicted as SUPERSEDE.

Unnecessary-abstention rate fraction of non-ABSTAIN items predicted as ABSTAIN. These make the cost of conservative errors visible.

Calibration and selective risk ten-bin top-label ECE, correctness Brier score, and accuracy/risk at coverage $0.25, 0.50, 0.75, 1.00$. These are descriptive without an independent calibration set.

Expected cost mean $C(a, y)$ under balanced, commit-averse, and update-responsive matrices declared in the released audit script.

Lineage integrity over gold-SUPERSEDE items, mean of a five-point row score: (1) old record marked `superseded`; (2) new record `active` with correct text; (3) new record's `supersedes_id` points at the old id; (4) old text no longer active in the slot; (5) exactly one active record of the new type in the slot.

Stale-retrieval rate over items declaring forbidden-answer patterns, the fraction whose predicted answer matches any forbidden pattern; lower is better.

Premise-resistance rate on premise-resistance items, the answer must include all required spans, exclude all forbidden spans, and contain an explicit rejection token.

Excluded `over_forgetting_rate` and `under_forgetting_rate` saturate at 1.0 for every system and never appear in a table.

Statistics. The primary interval is a 10,000-replicate cluster bootstrap over template families, repeated with minimal-pair groups and singleton unpaired items. Exact item-level McNemar is secondary. A zero action false-commit count uses the one-sided upper bound $1 - 0.05^{1/n}$. No equivalence or non-inferiority test is licensed because no margin was pre-specified.

E Reproducibility Commands

Validate the dataset.

```
python3 benchmarks/mgp6/validate_mgp6.py \
  --input benchmarks/mgp6/data/mgp6_v2.jsonl --strict
# expects: VALIDATION PASSED: 180 examples, 0 errors.
```

Reproduce the validity, cluster, calibration, and cost audit.

```
python3 scripts/analysis/audit_mgp6_validity.py
# writes artifacts/analysis/mgp6_validity_audit.{json,md}
```

Audit bibliography and citation coverage.

```
python3 scripts/release/audit_bibliography.py
```

Successor protocol. The released v3 design is prospective: it is not evidence that independent construction or three-human annotation has occurred.